

Hardware Guide for Running Nucleotide Transformer Models

Systems Compared

Desktop: Intel Core i5-12500, 32 GB DDR4 RAM, no dedicated GPU. Laptop: ASUS ROG Flow X13 with AMD Ryzen 9, 16 GB DDR5 RAM, and NVIDIA RTX 3050 (4 GB VRAM).

Best Use of the Desktop

Excellent for CPU-based inference with all 500M-parameter Nucleotide Transformer models, including NT-Human-Ref and NT-1000G base. It can also run 2.5B models for inference, but slowly and with high memory usage. Ideal for long-running jobs, bulk processing, and occasional large-model evaluation.

Best Use of the Laptop

The RTX 3050 dramatically accelerates inference for 500M models. It is the preferred system for interactive experimentation, embedding extraction, and rapid iteration. Sequence lengths up to about 1–2 kb are very comfortable on GPU, depending on memory usage.

Model Recommendations

500M models: Run on the laptop whenever possible for speed. 2.5B models (NT-1000G and NT-Multispecies): Run on the desktop using CPU and system RAM. Use the 500M models for screening and routine analyses; reserve 2.5B models for final validation or high-value sequences.

Expected Inference Times

500M model on desktop CPU (1 kb): approximately 2–4 seconds per sequence. 500M model on RTX 3050 GPU (1 kb): approximately 0.2–0.5 seconds per sequence. 2.5B model on desktop CPU (1 kb): approximately 15–30 seconds per sequence.

Memory Considerations

Desktop: 32 GB RAM is ample for 500M models and adequate for 2.5B inference. Laptop: 16 GB RAM plus 4 GB VRAM is well suited for 500M models, but insufficient for practical 2.5B GPU inference.

Final Recommendation

Use the laptop for day-to-day work with 500M models because the GPU offers major speed advantages. Use the desktop for memory-intensive workloads and for occasional inference with the 2.5B models. Together, these systems form an excellent complementary setup for genomic foundation model research.